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Original Research

SynthMedic: Utilizing large language models for synthetic discharge summary generation, correction and validation☆

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ABSTRACT

Background and Objective: Synthetic clinical texts can improve transparency and reduce bias and costs when training and evaluating specialized language models in the medical domain. Synthetic texts are freely shareable, as they contain no real patient information, and can be customized for a specific task. The objective of this study is to develop a methodology for generating, validating, and correcting synthetic discharge summaries using LLMs without requiring any real patient data.

Methods: The proposed approach uses an LLM to generate synthetic discharge summaries for specific diseases and standard medical references from Merck Manuals to ground the generation in internationally accepted medical practices. We validate the generated summaries using LLMs as well as by human expert validation. In addition, we propose a method for automatic correction of the generated discharge summaries using Knowledge Graphs to ensure medical factual correctness.

Results: The conducted human expert evaluation shows that the generated synthetic discharge summaries are credible and factually accurate when provided with the medical reference context. The generated summaries achieve a System Usability Score of 94.35% based on a comprehensive rubric evaluated by medical professionals and a score of 93.65% on the Faithfulness metric evaluated by an LLM.

Conclusions: The proposed methodology can be utilized to generate high-quality synthetic discharge summaries for various diseases. The generated synthetic corpus consists of 900 discharge summaries in English representing nine socially significant diseases and is publicly available under an open license. The community can take advantage of the corpus and proposed methodology to train complex machine learning models, helping medical professionals in their daily work without using real patient data.

1. Introduction

The applicability of modern natural language processing methods (NLP) in the medical field depends on the availability of clinical text corpora, as training deep neural models requires a substantial amount of data. Creating a clinical corpus with real patient data is expensive and requires significant effort from medical professionals. Publishing such a corpus is restricted by the sensitive nature of the information and the corresponding legal regulations and ethical standards. Synthetic clinical texts are freely shareable, as they contain no real patient information, making them a viable alternative. Synthetic text generation does not involve medical professionals, and the size of the resulting corpus depends solely on the specific requirements of the downstream task. Furthermore, synthetic patient cases for rare diseases can be generated, reducing the bias towards more prevalent conditions [1].

This paper describes a method for using a large language model (LLM) to generate synthetic discharge summaries based on established medical practice in English, without relying on real patient discharge

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[1] https://github.com/ggrazh/synthetic-clinical-corpus/.

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summaries. Since factual correctness is of utmost importance in the medical domain, we also propose an approach that utilizes an LLM and knowledge graphs (KGs) to evaluate and correct the generated texts automatically. A team of medical professionals assessed the quality of the synthetic discharge summaries against a comprehensive rubric. All prompts, queries, and KGs are published in GitHub.[1]

Statement of significance	
Summary	Description
Problem or issue	Data scarcity and privacy risks are limiting factors for training clinical machine learning models with real patient data.
What is already known	LLMs are used for generating synthetic datasets, however, they are known for hallucinations. Ensuring LLM-generated content complies with medical standards is an important prerequisite for using it in the clinical domain.
What this paper adds	This study proposes a methodology for generating, validating, and correcting synthetic discharge summaries using LLMs tailored to specific diseases. To illustrate our method, we create a synthetic corpus of 900 records representing nine socially significant diseases.
Who would benefit from the knowledge in this paper	NLP researchers working in the clinical domain would benefit from the corpus creation methodology and will be empowered to train models, which in turn improve the everyday work of medical professionals.

2. Related work

2.1. Synthetic clinical data generation

The task of generating synthetic discharge summaries is relatively unexplored in the literature. LLMs have become increasingly prevalent for creating synthetic data, including clinical notes, as they can follow natural language instructions and generate text without examples. The authors of [2] prompt GPT-3.5-turbo to generate synthetic discharge summaries using the descriptions of ICD-10[2] codes found in anonymized discharge summaries of real patient cases from the MIMIC-IV dataset [3]. Many works focus on the generation of synthetic clinical notes using deep neural models [4–6]. Melamud et al. train a two-layer long short-term memory (LSTM) on deidentified clinical notes from the MIMIC-III dataset. Li et al. explore the capabilities of recurrent neural networks (RNN), generative adversarial networks (GAN), and transformer models to generate clinical notes based on de-identified discharge summaries from the i2b2-2010 challenge dataset [7]. Kweon et al. prompt GPT-3.5-turbo to generate synthetic clinical notes using deidentified patient cases from the PMC-Patients dataset [8]. The authors used synthetic notes to train an LLM, demonstrating no significant differences in performance between models trained on synthetic data and those trained on real patient data.

2.2. Synthetic clinical data validation

Generally, the automatic semantic evaluation of clinical text involves fact-checking a document (such as a synthetic discharge summary) against some ground-truth text (usually human-written). Abacha et al. compare several automatic metrics to evaluate the factual accuracy of clinical notes generated by an LLM based on doctor-patient conversation notes [9]. Four aspects of synthetic texts are assessed: significant factual errors, hallucinations, correct facts that are present in both generated and reference texts, and incorrect facts in the generated text (excluding hallucinations), such as inappropriate medication dosages and incorrect patient birth dates. The authors note that metrics such as BERTScore [10] and BARTScore [11], which use a language model, strongly correlate with expert evaluation of hallucinations in generated texts [9]. In addition, metrics that calculate the similarity between generated and reference texts based on the occurrence of medical terms strongly correlate with expert assessment of the number of correct facts in the generated text.

LLMs are increasingly used as the basis for automatic evaluation frameworks in various domains due to their large context window and the ability to 'learn' the evaluation task from the instructions provided [12–14]. An intuitive approach is to prompt the model to give a score for the evaluated text according to a fixed scale (e.g. 0–100) [15]. Interestingly, this produces results that correlate well with human preferences. However, the scores are sensitive to the design of the prompt [16]. Es et al. propose an alternative metric, faithfulness, which measures the extent to which a generated text adheres to the information in a reference text. The faithfulness score is calculated by first prompting an LLM to extract statements from the generated text. Then, an LLM is prompted to determine if each statement can be inferred from the context. The final faithfulness score for the evaluated text is the number of statements supported by the context divided by the total number of statements.

KGs are an intuitive approach to validating a text, as they provide the information in it as structured data. Wang et al. [17] tackle knowledge graph validation by ranking the probability of the correctness of graph facts, rejecting those with low probabilities. The proposed CrossVal algorithm uses an externally verified knowledge graph to generate incorrect samples by crossing it with the validated graph. LLMs are shown to be useful in KG-related tasks. In [18], the authors explore the performance of LLMs in tasks related to the construction and refinement of knowledge graphs, such as named entity recognition, relation extraction, and link prediction between concepts. LLMs have also been shown to help enrich an existing ontology [19]. Likewise, in [20], the GPT-4 model handles various tasks, including generating SPARQL queries on Wikidata, creating schema-based example graphs, detecting syntax errors in TTL files, and other tasks.

3. Methods

This section describes the data used in the study and the proposed methodology for the automated generation, validation, and correction of synthetic discharge summaries.

3.1. Data sources

Medical Reference - MSD (Merck) Manuals: The MSD (Merck) Manuals[3] is a comprehensive source of medical information for medical professionals, covering more than 2800 topics, including disorders, tests, diagnoses, and drugs. These manuals serve as a reference for various medical practice standards worldwide. They are updated regularly by an multidisciplinary team of medical specialists and are available under an open license. The MSD Professional Manual is a publicly accessible, semi-structured medical reference manual created by medical experts, containing up-to-date information on a variety of diseases. Its information is in English, allowing wide use, and is based on the latest recommendations, which are listed in the references for each disease. Each document contains between 4 and 10 sections that describe the disease, its symptoms and signs, as well as the diagnosis and treatment process. The average number of tokens per section is

[2] https://www.icd10data.com/.

[3] https://www.msdmanuals.com/professional (retrieved on 1 May 2024).

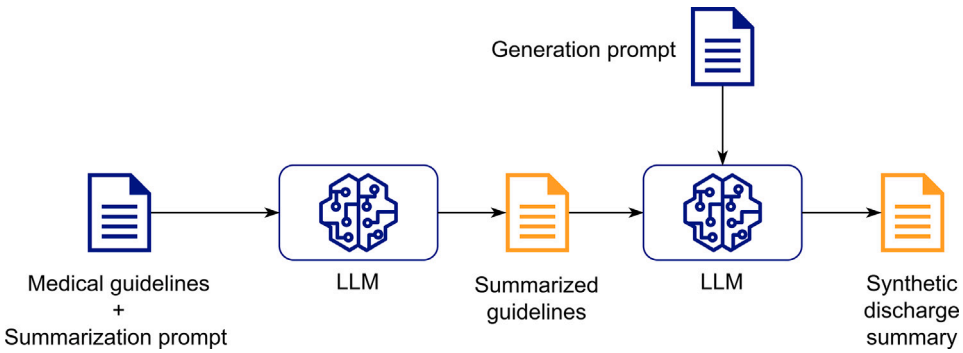


Fig. 1. The process of generating a synthetic discharge summary. A large language model first summarizes the medical information associated with the given diagnosis. The result is then used to generate the discharge summary.

660 (using OpenAI's tiktoken tokenizer[4]). We summarize the diagnosis and treatment sections of the reference text and use them to ground the discharge summary generation process, providing the LLM with the necessary information to generate a realistic diagnosis and treatment plan. Meanwhile, the automated evaluation process relies on the full text of the diagnosis and treatment sections to verify the facts. We pre-process the reference text by removing bibliographic references from the diagnosis and treatment sections to avoid generating incorrect verification rules that reference them. We also remove figure captions, as the models do not process images.

In this study, we sourced a limited number of short excerpts from the MSD manuals and used them as part of LLM prompts to guide the LLM's behavior. These excerpts were used solely to guide the generation of medically grounded discharge summaries and were not used for fine-tuning or redistributing the source content. The use of these excerpts was for research purposes only. No full-document content from the MSD manuals was stored, reproduced, or redistributed in the model outputs or dataset. The proper attribution has been maintained.

3.2. Medical diagnoses selection

To explore the applicability of the proposed methodology for generating discharge summaries, we selected some of the most prevalent and socially significant diagnoses, including *arterial hypertension (cardiology), hypothyroidism (endocrinology), diabetes mellitus type 1 (endocrinology), diabetes mellitus type 2 (endocrinology), influenza (infectious diseases), intracerebral hemorrhage (neurology), stroke (neurology), ankle fracture (traumatology), distal humeral fractures (traumatology)*. The proposed methodology is not limited to the listed diagnoses and medical areas. Some diagnoses, such as arterial hypertension and diabetes, are significantly more common in the literature and on the Internet. Consequently, they are well represented in the training corpora of popular LLMs.

The reference information we used for this study aligns with the latest international recommendations found in medical literature (Consensus statement for Type 2 Diabetes - [21]; European Guidelines for Stroke - [22,23]; Guidance for management of hypertension - [24]; Clinical guidelines for infectious disease - [25]; Guidelines for ankle fractures and Management of distal humerus fractures - [26,27]).

3.3. Synthetic discharge summaries generation

In a two-step process, an LLM is prompted to generate a synthetic discharge summary for a given diagnosis based on the respective medical reference document. The medical information is first summarized and then included in the instructions for an LLM to generate a discharge summary. Fig. 1 shows the generation process.

Step 1 - summarizing the medical reference text. On average, the Diagnostics and Treatment sections of the medical manuals have a combined length of about 2500 tokens (tiktoken tokenizer). Some of the language models we experimented with have a context window of 4096, shared between the prompt and the response, which significantly limits the length of the generated synthetic discharge summaries. For the method to apply to smaller models and to allow comparison between different model sizes, the medical reference text are summarized using an LLM. The summarization is done in three steps. First, an LLM is prompted to extract the key facts from the Diagnostics and Treatment sections. Then, an LLM provides feedback on the resulting summary, using the original reference as the source of truth. The feedback step focuses on the completeness of the summary, specifically whether key information is missing, such as the clinical presentation, diagnostic approach, names of drugs and their dosages, side effects, and discharge instructions. It also highlights parts of the summary that are not supported by the original text. Finally, the model is instructed to update the summary according to the feedback. The three steps were part of a single 'conversation', so the LLM has the original guidelines and different versions of the summary in its context window. The generation process is iterative, following the self-refinement procedure described by Madaan et al. [28]. The feedback generation and correction steps are repeated until there is no feedback to be addressed or a fixed number of iterations have been performed. Fig. 2 shows the summarization process. We developed the summarization prompt iteratively, following best practices in prompt engineering and analyzing the results empirically. It begins by specifying the role (the AI assistant of a medical doctor), which is important for producing quality summaries, as noted by Zheng et al. [29]. Some of the key information from the reference is related to the adverse effects of drugs, dosages, and contraindications, so we emphasize it in both the initial and feedback steps. For the final step, when the summary is updated to address the feedback, we note the importance of explicitly preserving statements that are not affected by the feedback. If this part of the prompt is omitted, the model creates a new summary, which could introduce errors in previously correct statements.

Step 2 - Generating the discharge summary To generate a synthetic discharge summary for a given diagnosis, a language model is prompted with the summarized text of the Diagnostics and Treatment sections of the corresponding medical reference. The process is illustrated in Fig. 1. The model should select the appropriate information from the reference text to use during generation. For example, when generating a discharge summary for a patient with type 1 diabetes, the model is expected to select only the relevant parts for that diagnosis from the reference text, which includes information on both type 1 and type 2 diabetes.

[4] https://github.com/openai/tiktoken.

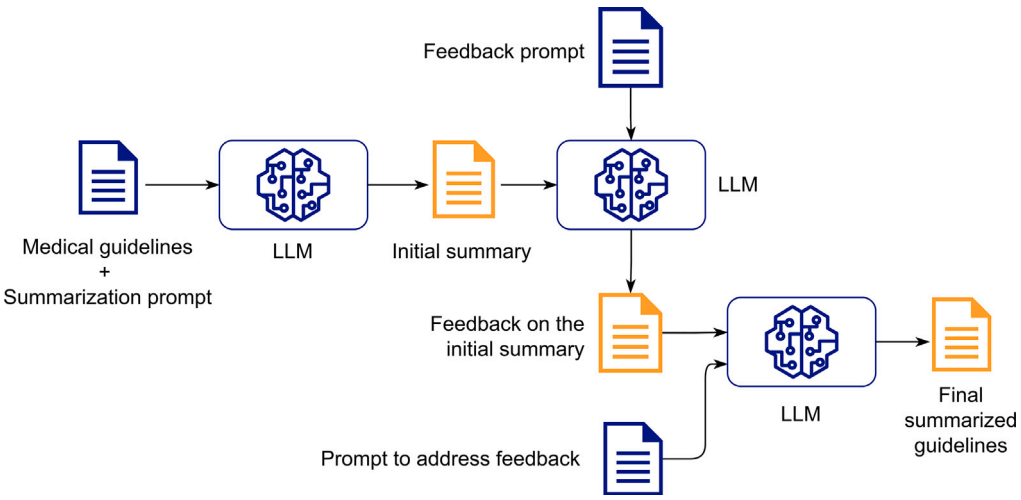


Fig. 2. The medical reference summarization process. A large language model first summarizes the text from the Diagnostics and Treatment sections of the reference text. Then, the summary is iteratively refined and updated.

3.4. Synthetic discharge summaries feedback and validation

3.4.1. Feedback and validation using LLM

The proposed validation method features a sequence of instructions for an LLM to calculate the faithfulness score of a synthetic discharge summary. Faithfulness is a metric used to evaluate RAG-based systems, where there is a question, some context extracted by the RAG system, and an answer. Faithfulness assesses the extent to which the statements in the response to a question are supported by the context of the question [16]. The complete validation flow is shown in Fig. 3.

To calculate the faithfulness score of a synthetic discharge summary, we first generate statements based on the sections 'Diagnosis' and 'Treatment' of the medical reference for the given disease using an LLM. We then verify whether these statements can be inferred from the synthetic discharge summary by prompting the LLM. Faithfulness is calculated as the number of statements supported by the synthetic discharge summary over the total number of statements for the specific disease as shown in Eq. (1).

$$\text{Faithfulness} = \frac{\text{num. statements inferrable from the discharge summary}}{\text{total num. statements for disease}} \tag{1}$$

The validation process consists of four steps:

1. Statement generation — Using an LLM, we generate statements from the Diagnosis and Treatment sections of the medical reference for a given disease.
2. Filtering statements that depend on the context — we prompt an LLM to classify the statements as either context-dependent (conditional) or context-independent (concrete facts).
3. Filtering relevant statements for a specific discharge summary — using an LLM, we determine which of the context-dependent statements are relevant in the context of a particular discharge summary.
4. Calculating the faithfulness of a discharge summary, we combine all context-independent statements with those identified as relevant for the given discharge summary. The resulting list of statements is used to calculate the faithfulness of the discharge summary. An LLM is prompted with the statements and the text of the discharge summary to determine which statements are supported by the context of the discharge summary.

Step 1 - Statements generation. In the first step of the validation process, we generate a list of statements for each disease. The statements should cover all aspects of the diagnosis and treatment of a disease, which are described in the medical reference. To compile the list, we first prompt the LLM to generate an initial set of statements. In the same chat session, we then instruct the model to review the reference once more and revise the list to add any statements that may have been missed. This follows the self-refinement approach described by Madaan et al. [28]. The prompt instructs the model to add any missing statements and fix any vague or incomplete statements. We use a few-shot prompt to generate the statements, specifying the role of the model (in this case, "a medical doctor's AI assistant"). There is also an explicit requirement that the generated statements are based solely on the information in the provided reference, emphasizing the importance of details related to medication dosages and time constraints (for instance, in emergency cases, several procedures must be performed within the first three hours of the onset of specific symptoms). The prompt includes multiple examples to guide the model towards a unified format: statements without conditions use "must" instead of "should", while conditional statements always begin with the condition. Each example in the prompt includes text from the medical reference and an expected statement derived from that text. Formulating the statements using the more assertive "must" plays an important role in subsequent evaluation steps, as it conditions the model to treat statements without conditions as requirements rather than optional recommendations. Based on an ad-hoc test using a few examples, we observed that this reduces the number of statements falsely classified as supported by the context, as the model considers them optional. In our limited test, we noticed that formulating non-conditional statements with the more assertive "must" can positively impact the accuracy of subsequent evaluation steps. Specifically, using "must" prompts the model to interpret such statements as requirements rather than optional recommendations, which seems to reduce the frequency of statements incorrectly classified as supported by the context based on ad-hoc testing. This approach aligns with the nature of the medical reference, which uses the word "should" to describe generally accepted diagnostic and treatment algorithms that convey an intermediate level of obligation between "must" (highest obligation) and "may" (lowest obligation) [30]. This observation is consistent with prior work demonstrating that large language models are sensitive to the choice of epistemic modals (e.g., "must", "may", "might") in input prompts [31]. Finally, ensuring that conditional statements adhere to the standard "if, then" format seems to improve accuracy when determining whether a given statement is supported by the discharge summary context based on a few ad-hoc tests.

Step 2 - Filtering statements dependent on the context. Once the validation statements for a specific diagnosis are generated, we noticed

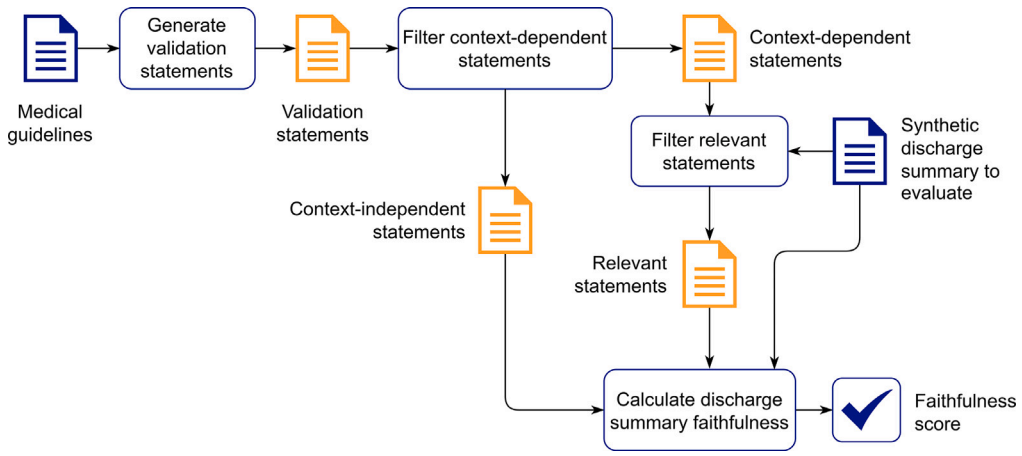


Fig. 3. The process of calculating the faithfulness of a synthetic discharge summary. First, validation statements are generated based on the medical reference for the corresponding diagnosis. Then, the statements are filtered in two steps to compile the final validation list, which is used to calculate the faithfulness of the given discharge summary.

that some of them do not contain any conditions and are therefore independent of the discharge summary context. These statements are always verified against any discharge summary for the given diagnosis and do not require further processing, so they are filtered using specific LLM prompt instructions. We define each of the two classes of statements (dependent and independent) and provide examples in JSON format. The "explanation" field in each example allows us to define the classes further. Although it is also included in the model's answer, we ignore it during the validation process. This step is performed once for each diagnosis, and the resulting list of independent statements can be reused.

Step 3 - Filtering relevant statements for a discharge summary. When evaluating a particular synthetic discharge summary, we must determine which of the dependent statements from the previous step should be verified. If the discharge summary meets the predicate of a conditional statement, that statement is included in the final validation list.

The instructions used to classify the dependent statements as relevant or irrelevant specify the role of the model and the context of the task. Including examples in the prompt is crucial. Of special importance are examples where the predicate of the statement is fulfilled, but the discharge summary does not entail the statement as a whole. The LLM often incorrectly classifies such examples as irrelevant, even though they highlight key mistakes in the evaluated texts.

Step 4 - Calculating the faithfulness of a discharge summary. In the final step of the validation process, we calculate the faithfulness score for the synthetic discharge summary. At this point, we have the text of the discharge summary and a list of statements:

- independent of the context
- dependent on the context and relevant for the given discharge summary

We prompt an LLM to determine whether each of the statements can be entailed from the context of the discharge summary. The final faithfulness score is calculated as the number of entailed statements over the total number of statements in the evaluation list. A key benefit of this approach is that, in the end, we not only know the faithfulness score but also which statements contradicted the discharge summary context, allowing us to use them in the automated improvement process. Similarly to the other steps, they specify the role of the model, describe the task, and provide a set of examples of statements that can be entailed, as well as examples of statements that contradict the discharge summary.

Even though the Faithfulness metric is becoming more commonly used for evaluating whether the generated text is grounded in the source material, it relies on the capabilities of the evaluating LLM, which poses concerns about potential biases and hallucinations [16,32,33]. LLM evaluations also struggle with complex reasoning chains and long premise length [34]. Thus, we instruct the model to generate simpler atomic statements for validation that will be easier for the LLM to evaluate.

Due to the critical nature of clinical NLP tasks, we employ a strict definition of faithfulness: a validation statement is considered contradicted if the generated discharge summary does not fully support it. This approach differs from popular evaluation frameworks, which typically assign a score (e.g., 1 to 5) [32] to each statement to indicate varying degrees of faithfulness. The faithfulness metric focuses solely on the correctness of diagnostic and treatment information in the evaluation discharge summary. Additionally, it does not account for any potentially redundant information in the generated discharge summary.

3.4.2. Feedback and validation using LLM and KGs

To validate the correctness of the discharge summary, we use the Merck Manuals description of the primary diagnosis as a disease knowledge source. First, we generate a knowledge graph representing each disease's medical reference using an LLM (GPT-4). Then, using the disease KG, we follow a multi-step process to generate a knowledge graph from the discharge summary. Using the generated structured representations, the system generates feedback on incorrect statements in the discharge summary. Finally, using the generated feedback, we calculate the faithfulness metric as defined in the previous section. Below we describe each step in more detail.

Generating knowledge graph from the disease medical reference. To generate a knowledge graph from the disease medical reference, we use the full description of the disease from the Merck Manuals, splitting it into smaller chunks. We prompt an LLM for each chunk to generate a knowledge graph representing the information in the reference text using the HL7 FHIR[5] schema and the Turtle format (TTL).[6] Next, the LLM is prompted to validate the completeness and correctness of the knowledge graph, and finally to generate an improved version. In some edge cases, the TTL format generated by the LLM is not valid, so if parsing is not possible, the system skips that chunk. Therefore, the final KG may be incomplete in terms of covering all entities and relationships mentioned in the text.

[5] https://www.hl7.org/fhir/.
[6] https://www.w3.org/TR/turtle/.

We merge the graphs for each chunk and produce a single knowledge graph that represents the whole reference text. We used rdflib[7] to parse the LLM-generated TTL representation of the subgraph and validate the result, then combine the TTL for all subgraphs. We use deduplication methods to combine duplicate entities from the resulting knowledge graph. We use the library dedupe, which clusters nodes in the graph and finds and combines duplicate entities using similarity metrics.[8] The KG is bi-directional, as it represents different predicate types that correspond to various relationships between entities. For example, Stroke is a related condition to Ischemic Stroke, and Ischemic Stroke is related to Stroke. Cycles may exist in such a graph, as it represents multiple relationships between the entities. The generated knowledge graph contains codes from common medical ontologies and standards, such as SNOMED, LOINC, and RxNorm; however, these codes are often incorrect. Therefore, we perform a post-processing step to correct these codes by using the mention text extracted from the reference text. We utilize the GPT text-embedding-3-small model[9] to generate embeddings for the reference text mentions and all concepts from the medical ontologies. Using the generated embeddings, we perform an offline cosine similarity search, select the closest concept from the ontology, and replace the code in the knowledge graph.

The knowledge graph is further enriched with data from DrugBank[10] [35] related to the dose and route of administration of medication. DrugBank is a database of medications, providing information on over 17,000 drugs. It has a free license for access for academic research, which is the case in this study. For each disease mentioned in the reference text, the knowledge graph is enriched with the DrugBank information for drugs, dosage, route, form, and strength. This graph will be referred to as a disease knowledge graph and will serve as the source of truth for medical reference information in the validation of the discharge summary.

The reference ontologies and systems, such as SNOMED, DrugBank, LOINC, and RxNORM, were used in offline validation and graph enrichment, but not in LLM prompting, reproduction, or redistribution in any way.

Generating knowledge graph from discharge summary. Similar to the disease text, the long text of the discharge summaries led to the loss of some of their context while generating and validating the knowledge graphs. Therefore, we split the text into smaller chunks, and a knowledge graph is generated and validated for each chunk.

For the discharge summary KG generation task, we used OpenAI's functionality for assistants and threads.[11] This allowed defining an assistant for each step of the process. Assistants can execute functions during their invocation. Each of the defined assistants is instructed to generate SPARQL queries to retrieve necessary information from the disease knowledge graph during the process of generating the discharge summary knowledge graph. The instructions for each assistant are divided into three parts: utilizing SPARQL queries, a description of the task the assistant will perform, and a description of the data. These instructions are concatenated and provided to the assistant.

The graph generation approach is divided into several steps:

- Generating an initial version of the graph.
- Improving the initial version of the graph.
- Adding missing concepts and relationships to the graph through a feedback mechanism.

Each of these steps is handled by a separate assistant as shown in Fig. 4.

Step 1 - Generating the initial version of the KG. To generate the initial version of the graph, the model is instructed to follow several steps — identify all concepts in the text, identify the relationships between the discovered concepts, extract information about the discovered concepts from the provided disease knowledge graph, review what can be reused, and generate the results from the previous steps in TTL format. The instructions follow prompt engineering best practices by specifying the assistant's role and breaking down the process into multiple steps. In addition, the target schema for graph generation is specified to be the HL7 FHIR standard. Ultimately, the assistant is prompted to generate an rdfs:label property for each identified concept and to refrain from utilizing knowledge from the disease graph.

Step 2 - Improving the initial version of the KG In the second step, a different assistant is applied, which is used to provide explanations and suggestions on how to improve the graph. The goal is to improve the KG in the following three aspects:

- Completeness — check if there are missing objects from the text.
- Validity — ensure compliance with the HL7 FHIR schema.
- Fidelity to the text — remove statements that cannot be traced back to specific information in the text.

After applying these comments, the assistant generates an improved version of the graph.

Step 3 - Identifying concepts and relations using external models. To identify information from the text missing in the generated KG, we utilize smaller Named Entity Recognition (NER) and Relation Extraction (RE) models, which are publicly available. To extract named entities from the text, we use Biomedical Stanza.[12] For the relation extraction task, we use OpenIE,[13] which generates the relations as a triplet of subject, predicate, and object, making the results easy to integrate into any knowledge graph. As a result, we have a list of concepts and relations extracted from the discharge summary.

Step 4 - Adding any missing concepts and relationships. Using the results of Step 3, we compile a list of concepts and relations that appear in the discharge summary but are missing in the KG generated by LLM. We use the rdfs:label property to compare and identify missing concept names. The list of missing concepts and relations is provided as a user instruction in the assistant's thread, with the task of incorporating the missing elements into the knowledge graph. This step can be repeated several times until there are no more missing concepts or relations, or until the maximum number of iterations has been reached. Finally, when the process is complete, KG is generated that represents the discharge summary.

Validation using the Generated KG. Given both the knowledge graph, generated from the discharge summary (referred to as the discharge summary knowledge graph), and the disease graph, generated from the medical reference text, another GPT assistant is prompted to validate the information in the discharge summary graph and produce a list of inconsistent statements. The assistant is instructed on how to identify discrepancies in the discharge summary graph by being provided with a well-defined explanation of the task, the relevant SPARQL queries to execute, and examples of inconsistent statements. The instruction provides examples to help guide the assistant towards the types of errors to look for and an example of the structure for each detected inconsistency. In addition, the model is provided with all the data from the disease knowledge graph as well as the discharge summary text. As a result, the LLM generates a list of inconsistencies between the discharge summary and the disease graphs, along with an explanation for each result.

3.4.3. Feedback and validation by human experts

Human expert evaluation of any automatically generated text is essential to ensure the reliability of the information [36]. This is especially critical when developing a synthetic clinical corpus, as any

[7] https://rdflib.readthedocs.io/.
[8] https://dedupe.io/.
[9] https://platform.openai.com/docs/models/text-embedding-3-small.
[10] https://go.drugbank.com/.
[11] https://platform.openai.com/docs/assistants/overview.
[12] https://stanfordnlp.github.io/stanza/available_biomed_models.html.
[13] https://stanfordnlp.github.io/CoreNLP/openie.html.

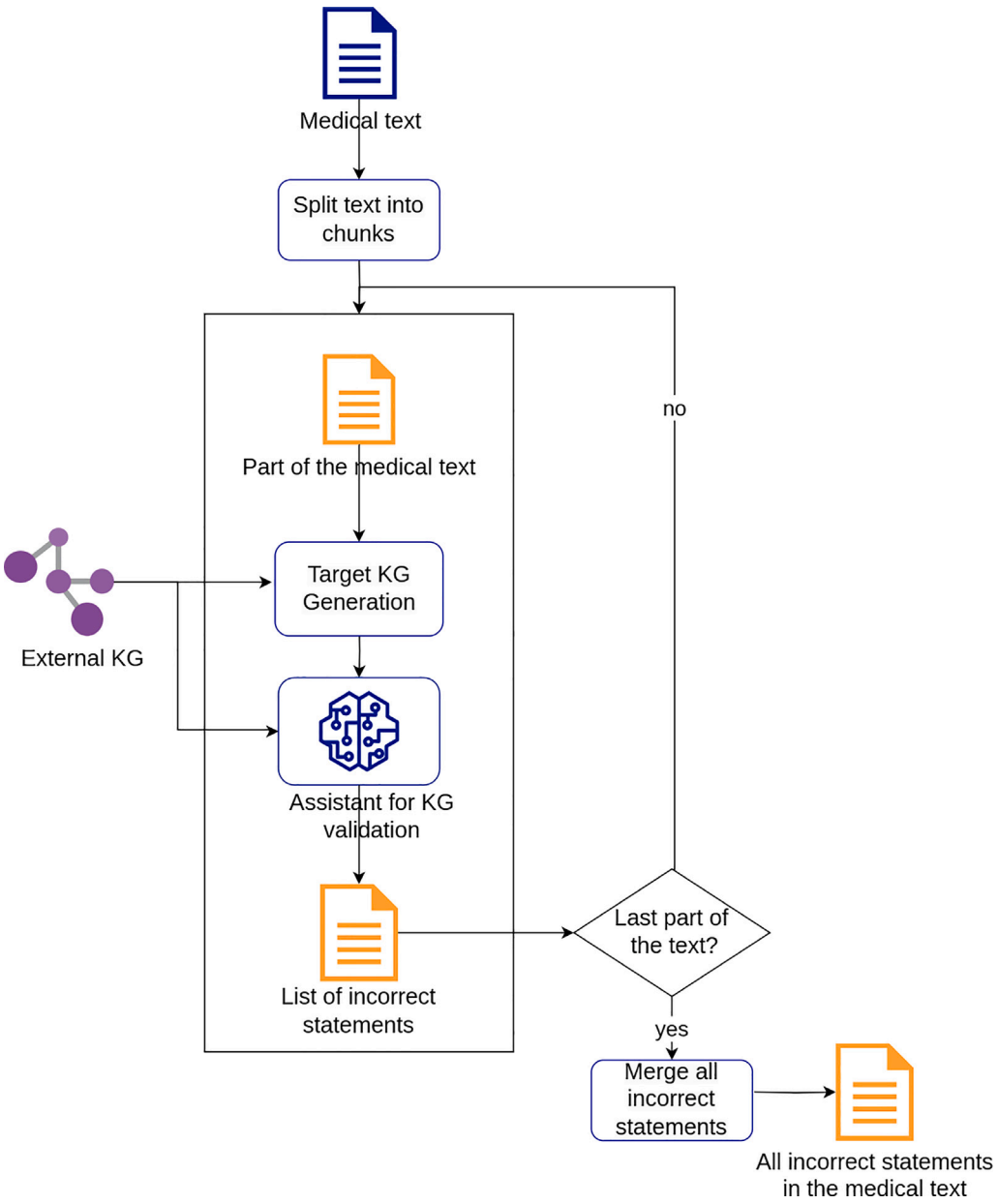


Fig. 4. The process of generating a KG from a discharge summary. First, an initial version is generated using an OpenAI assistant. Then, it is improved by a second assistant. Finally, a third assistant is used to introduce statements in the discharge summary, utilizing feedback data from KG.

inaccuracies within the corpus can adversely affect model performance and credibility in downstream tasks.

Two neurologists evaluated the quality of 103 synthetic discharge summaries generated using GPT-4-0125 and Llama 3 8B Instruct. The format of the documents produced by both models is similar to minimize evaluator bias towards a particular model. The summaries represent two diagnoses, intracerebral hemorrhage (55 summaries) and ischemic stroke (48 summaries). Each reviewer is assigned to review all summaries for a single diagnosis, and each summary is evaluated only once. Experts assessed the accuracy, completeness, redundancy, and style of synthetic discharge summaries. The primary source of medical information is the diagnostic and treatment reference of the MSD manuals.[14] [15]

For each discharge summary, experts provided comments highlighting any inaccuracies identified during the evaluation process and completed a questionnaire assessing the quality of information in the synthetic text. The questionnaire was in the format of the System Usability Scale (SUS) [37] and consisted of 10 statements about the content of the synthetic discharge summary, alternating between positive and negative statements. For each statement, the medical specialist assigned a score on a scale from 1 (strongly disagree) to 5 (strongly agree). The final score for the System Usability Score (SUS) of the synthetic discharge summary was calculated using the following formula presented in Eq. (2).

$$\mathrm{SUS} = 2.5((P - 5) + (25 - N)) \tag{2}$$

[14] https://www.msdmanuals.com/professional/neurologic-disorders/stroke/intracerebral-hemorrhage (retrieved on 1 May 2024).

[15] https://www.msdmanuals.com/professional/neurologic-disorders/stroke/ischemic-stroke (Retrieved on May 1, 2024).

where P is the sum of scores of positive statements, and N is the sum of scores of negative statements. The score on this scale ranges from 0 to 100. A discharge summary with a score above 80 is considered acceptable.

3.5. Synthetic discharge summaries correction and revalidation

After the validation and feedback system processes a given synthetic discharge summary, it produces as feedback a list of recommendations to improve the text. These recommendations are in natural language and do not adhere to a fixed format. They can address various quality aspects, such as the accuracy, completeness, and redundancy of information, as well as the document structure and style. During the automatic correction step, these recommendations are applied to improve the synthetic clinical text. A significant challenge is to make the recommended changes without affecting parts of the synthetic text that are already correct. To achieve this, we prompt an LLM in two stages:

1. For each recommendation, we generate a series of atomic editing operations and identify the section of the text where each operation should be applied. The supported operations are adding, deleting, and updating text.
2. We then prompt an LLM with the list of editing operations along with the current version of the synthetic text. The model executes the sequence of editing operations in the corresponding sections of the text.

Using a specific instruction prompt, we use the LLM to generate a sequence of steps (plan) to edit a given synthetic discharge summary. They define three primary operations: adding text, deleting text, and replacing text. Each feedback item can be expressed using one or more of these operations. Similarly to the other prompts, the model's role is specified to set the context of the task. Next, the three types of operations are defined, along with examples. In addition to the feedback text, the prompt also includes the synthetic discharge summary, allowing the model to determine the specific section of the text where each operation should be applied.

The instructions for applying the edit plan aim to satisfy the following requirements.

- The operations must be performed sequentially, starting from the first. This guides the model in preserving the consistency of the text, as the order of the editing operations determines the result.
- No edits should be made that are not included in the list of operations. The model is instructed to keep existing content, which is not addressed in the feedback, unchanged. This approach reduces the number of errors caused by unnecessary paraphrasing of initially correct text sections.

After the automatic correction step is performed, we revalidate the corrected discharge summaries using LLM and human experts.

4. Results

In this section, we describe the process of creating and refining the instruction sets used to generate, validate, and correct synthetic discharge summaries. We also share our approach to automated and human evaluations and analyze the results.

4.1. Synthetic discharge summaries generation

Due to the prompt and input text length, we used large language models that support a long context. For summarizing medical reference texts, we used gpt-4-0125-preview [38]. For generating synthetic discharge summaries, we experimented with gpt-4-0125-preview and Llama 3 8B Instruct [39]. We performed initial experiments with Mistral 7B-Instruct v0.2 [40] and the biomedical model BioMistral 7B-DARE [41], but the resulting generated summaries were of lower quality — incomplete, or repeated the same sentence multiple times. Therefore, we decided to proceed with GPT-4 and Llama 3 as the primary models for generating the dataset.

The generated corpus consists of 900 synthetic discharge summaries representing nine socially significant diseases. We generated 50 summaries per disease using both LLMs: GPT-4 and Llama 3.8 B Instruct.

Table 1
Average system usability score (SUS) given by the human experts to the evaluated synthetic discharge summaries.

Diagnosis	Number of evaluated summaries	Average SUS
Intracerebral hemorrhage	55	94.40
Ischemic stroke	48	62.86

4.2. Synthetic discharge summaries feedback and validation

In this section, we present the results from the validation of the generation methodology and the synthetic corpus by LLM and human experts.

4.2.1. Feedback and validation using LLM

To assess model accuracy on the validation tasks, we evaluate each part of the system individually.

Relevant Statement Classification. The dataset for context-dependent relevant statement classification comprises 135 statements on 16 synthetic discharge summaries and is labeled by a non-expert. Using the proposed prompt to filter relevant and irrelevant statements, the GPT-4-0125-preview achieved 92% accuracy on the evaluation dataset. The most common error occured when the model attempts to verify the entire conditional statement against a discharge summary, rather than only the predicate. In some rare cases, the model incorrectly classified the statement as irrelevant even though it was completely relevant.

Faithfulness Entailment Classification The dataset for assessing faithfulness includes 203 statements on 24 discharge summaries labeled with contradiction/entailment and is labeled by a non-expert. GPT-4 performed well on the statement inference evaluation dataset, with an accuracy of 97%. The few (four) errors were on statements with multiple requirements where the model allows some of the requirements not to be explicitly met, assuming them from the context.

Overall, the generated summaries achieved a score of 93.65% on the Faithfulness metric evaluated by an LLM.

4.2.2. Feedback and validation by human experts

In this section, we analyze the results of the human expert evaluation of 103 synthetic discharge summaries. Table 1 displays the average scores given by medical professionals on the quality of information in synthetic discharge summaries for intracerebral hemorrhage and stroke. The average System Usability Score across all evaluated summaries is 94.35%.

There is a notable difference in key aspects of information quality between the two diagnoses, which may be due to ischemic stroke being one of the most common (and socially significant) diseases. There is significantly more publicly available information about it on the internet compared to intracerebral hemorrhage, and its quality varies. This noise likely negatively affects the reliability of the information in the synthetic discharge summaries for ischemic stroke.

According to experts, language models performed quite well in generating relevant symptoms in the synthetic discharge summaries for both diagnoses. There were gaps in the summary for ischemic stroke, where the average diagnostic rating was 3.90, due to missing important information, including the etiology of the stroke, blood pressure values

upon patient admission, and key diagnostic procedures such as a CT scan.

A significant difference was observed between the discharge summaries for the two diagnoses in the assessment of whether the treatment plan follows commonly accepted practices. In some cases of ischemic stroke, the language model incorrectly assessed the severity of the stroke and often prescribed a treatment plan suitable only for mild cases, which resulted in the use of inappropriate combinations of medications.

Regarding the evaluation of how appropriate the language was in the synthetic discharge summaries, the experts found that LLMs are capable of generating coherent and well-structured discharge summaries. The summaries for ischemic stroke showed a lower score for completeness of information (average of 3.20), primarily due to the omission of important details such as stroke etiology, risk factors, and time of admission. A small portion of the evaluated summaries included unnecessary information, describing which procedures were not performed, and which medications were not prescribed. According to the evaluating experts, this was not incorrect but was unnecessary in most cases — very rarely seen in real discharge summaries.

4.2.3. LLMs generation quality comparison

To assess the effect of different models on the quality of synthetic discharge summaries, we compared the human evaluation scores and the faithfulness scores of 56 intracerebral hemorrhage texts generated by GPT-4-0125-preview and Llama 3 8B Instruct, respectively. We found that GPT-4 achieved a higher average SUS score — 98.3%, compared to Llama's 90.4%. The GPT model also ranked higher on faithfulness, with a score of 98.40% versus 88.90% for Llama.

4.3. Synthetic discharge summaries correction and revalidation

We used a method for automatic correction based on LLM feedback to update the discharge summaries, and then performed a revalidation on a subset of the corrected summaries.

4.3.1. Feedback and revalidation using LLM and KGs

We used 88 discharge summaries to revalidate the automatic correction. The extracted disease KGs were utilized in the revalidation process to ground the LLM in the feedback generation process. Several incorrect statements were introduced in the discharge summaries to verify if the validation pipeline would find them. The validation pipeline outputs the incorrect statement from the discharge summary graph, the corresponding corrected statement, and an explanation of the error in the discharge summary graph, along with instructions on how to fix it. We evaluated the faithfulness of the discharge summary before and after applying the comments from the validation assistant. Fig. 5 shows the average faithfulness results grouped by diseases of the experiments with KG validation.

We evaluated the automatically corrected discharge summaries using the Faithfulness metric and compared them to the score of the originals. For the subset of 88 discharge summaries, the original Faithfulness was 70.70% on average, while the corrected summaries had an average score of 78.98% - an improvement of 11% based on the corrections.

4.3.2. Feedback and revalidation by human experts

We also performed a human evaluation of the corrected discharge summaries for a small randomized subset of the generated corpus. Physicians thoroughly checked the corrected synthetic texts. The revalidation by experts demonstrated that the language model performed the recommended edits with high accuracy — there were no inaccuracies resulting from the correction step, nor was any information deleted outside of the recommendations. About 66% of the updated discharge summaries were determined by doctors as excellent. The comments for the rest were related to new information that should be added and were not associated with the corrections made. For one of the discharge summaries, the specialists indicated a significant inaccuracy caused by the edits — the prescribed combination of medications was incorrect. The problem of detecting drug interactions is very complex and requires further study.

5. Discussion

5.1. Human expert vs. LLM validation

We compared the human evaluation SUS metric with the Faithfulness automated evaluation metric. We calculated the Faithfulness score for the 103 discharge summaries that underwent manual evaluation. For each diagnosis, we applied the Wilcoxon signed-rank test to determine whether there was a statistically significant discrepancy between automated and expert evaluations. In the case of intracerebral hemorrhage, the difference was not statistically significant (p-value: 0.856). The SUS score for intracerebral hemorrhage was 94.4%. However, for ischemic stroke cases, the difference between human and automated evaluation was statistically significant (p-value: 8.526e−14), and the SUS score was 62.86%. We attribute this difference to several different properties of the automatic Faithfulness metric and LLM evaluation:

- Faithfulness relies only on information from the diagnostic and treatment reference, while human experts use all their background knowledge in evaluations. Medical knowledge, such as specific drug names and default medication dosages, is not present in the reference text. For example, in the reference texts, when it comes to medications, classes of medications (such as beta-blockers, corticosteroids, and anticoagulants) are almost always mentioned instead of their commercial names. In some disease references, the dosage for taking some medications is not specified. In such cases, it is assumed that the default daily dosage (as described in the medication leaflet) is used. However, since the context does not include this information, the automatic metric cannot verify the correctness of the medication regimen. Medication dosage was one of the primary errors identified by human experts in ischemic stroke discharge summaries.
- The automatic metric does not check for redundant information — appointment time, redundant diagnostic procedures, etc. Almost all generated verification statements focus on the presence and correctness of certain information. Human experts have taken into account redundant information and adjusted their scores accordingly.
- LLM evaluation does not seem to consider values of specific numerical scales like National Institutes of Health Stroke Scale (NIHSS),[16] which determines the severity of the stroke. This is key in verification, as treatment varies dramatically depending on how a particular case of ischemic stroke is classified on the scale. There is no such scale for Intracerebral hemorrhage, so that could explain the difference between the disease scores.

In general, the automated faithfulness metric can be a good indicator of the quality of information in synthetic discharge summaries, provided that the evaluating LLM has access to reliable medical sources. Integrating multiple reference sources, each focusing on a specific quality aspect, can address some of the identified limitations.

5.2. Bias and diversity

5.2.1. Patient age distribution

We group synthetic discharge summaries by age into seven categories. The general age distribution is shown in Fig. 6a. For both

[16] https://www.ninds.nih.gov/health-information/stroke/assess-and-treat/nih-stroke-scale (Retrieved on October 1, 2024).

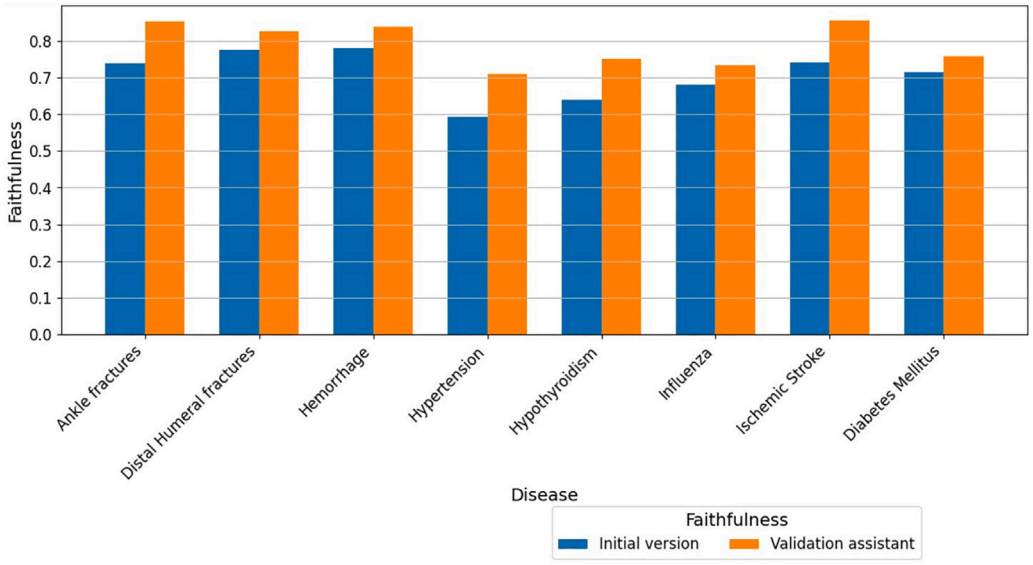


Fig. 5. Faithfulness scores of the discharge summaries before and after applying the comments from the validation with KGs.

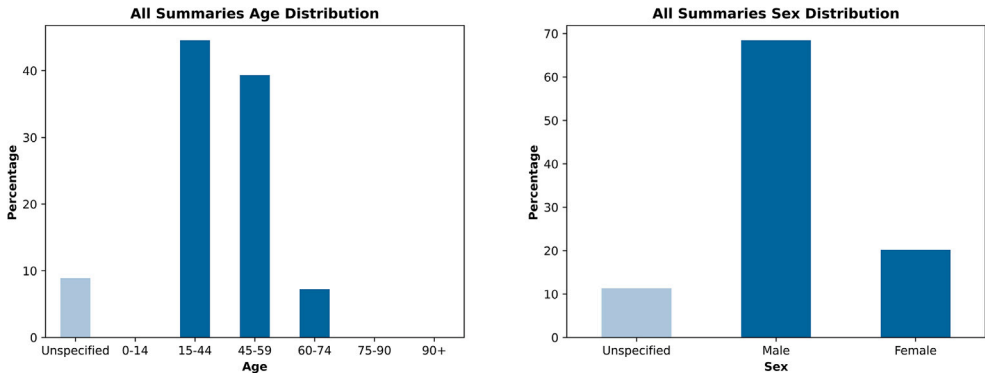


Fig. 6. The distribution of patient age (a) and sex (b) in the generated corpus of 900 synthetic discharge summaries.

diagnoses, the actual prevalence of the disease does not match the distribution generated by the model. In the case of intracerebral hemorrhage, approximately 65% of the generated instances are for patients between 45 and 59 years of age; however, in the US for the period 2004–2018, the predominant age group is over 75 years old [42]. In the case of ischemic stroke, the predominant age group in the synthetic data is 45–59 (50% of the texts). At the same time, the statistics for US patients show the prevalence of the 65+ group [43]. This discrepancy can be addressed by altering the generation prompt to specify the patient age.

5.2.2. Patient sex distribution

When the prompt does not explicitly specify the patient's sex assigned at birth, both LLMs tend to generate discharge summaries for male patients (almost 70% of the cases). Fig. 6b shows the distribution. Examining the data for specific diagnoses reveals a similar bias towards male patients, which does not align with the actual diagnosis statistics for the US [42,43].

To mitigate this bias towards male patients and younger age groups, explicitly setting the patient's sex and age can be included in the prompt when generating the discharge summary. As future work, updating the generation prompt and creating a more balanced dataset can be explored.

5.2.3. Procedures, diseases, drugs, and lab tests

The statistics on data diversity are presented in Table 2, which combines data for each model, GPT and Llama. The GPT model tends to produce more extensive and varied output in terms of total and unique occurrences, as it has higher mean and total values for each entity type.

Table 2
Statistics on unique and total occurrences of various entity types in the corpus. The 'All' section presents aggregated statistics from summaries generated by both the GPT and LLaMA models.

Data source	Statistic type	Diseases	Symptoms & signs	Treatments	Tests
All	Total count	3999	15 503	14 477	13 575
	Unique count	461	2027	2184	1984
	Mean count	4.44	17.23	16.09	15.08
	Mean unique count	3.09	11.79	12.36	12.69
GPT	Total count	2239	8951	8187	6689
	Unique count	288	1346	1439	1233
	Mean count	4.98	19.89	18.19	14.86
	Mean unique count	3.34	13.32	13.47	12.42
LLaMA	Total count	1760	6552	6290	6886
	Unique count	279	1030	1137	1172
	Mean count	3.91	14.56	13.98	15.30
	Mean unique count	2.85	10.26	11.25	12.97

Although the source reference page for Diabetes contains information about types 1 and 2, the LLMs appear to be able to select the correct part of the information based on the diagnosis set in the prompt. The human expert evaluation did not identify any confusion between the two types, which may be explained by the fact that the LLMs have

been trained on large amounts of data and can therefore differentiate between them.

6. Conclusion

We presented a methodology for automated generation, validation, and correction of synthetic discharge summaries using LLMs. The expert evaluation conducted shows that the generated texts generally adhere to the expected structure and content of real discharge summaries. Additionally, results from the automated evaluation metric, Faithfulness, align with expert preferences when the evaluating LLM is provided with the necessary context. The generated summaries achieved a System Usability Score of 94.35% based on a comprehensive rubric evaluation by medical professionals and a score of 93.65% on the Faithfulness metric evaluated by an LLM.

The proposed approach can be leveraged to generate high-quality synthetic discharge summaries for different diseases. The synthetic corpus we created consists of 900 discharge summaries in English representing nine socially significant diseases and can be used by the community to train machine learning models for the clinical domain.

Future work can address some of the limitations by integrating multiple sources that cover various aspects of diagnosing and treating a particular disease and incorporating a secondary evaluation system that models drug interactions and doses. Expanding the proposed methodology to languages other than English would be beneficial, as synthetic clinical texts are particularly valuable for low-resource languages.

CRediT authorship contribution statement

Georgi Grazhdanski: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vasil Vasilev:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sylvia Vassileva:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Dimitar Taskov:** Writing – review & editing, Writing – original draft, Validation, Data curation. **Izabel Antova:** Writing – review & editing, Writing – original draft, Validation, Data curation. **Ivan Koychev:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Svetla Boytcheva:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Limitations

The proposed approach for generating, validating, and correcting synthetic discharge summaries has several limitations. First, although the Faithfulness metric is becoming increasingly common, it relies on the capabilities of the evaluating LLM, which raises concerns about potential biases and hallucinations. To minimize the risk of factually incorrect results, the described automated methods require the use of some of the largest language models, which negatively impact system throughput and the cost of verifying a single synthetic discharge summary. In addition, the faithfulness metric focuses solely on the correctness of diagnostic and treatment information in the evaluation discharge summary. Also, it does not account for any potentially redundant information in the generated discharge summary. Additional metrics are required to assess the dosages of the medications and the style of the clinical document. Lastly, in the context of automatic correction, it is worth noting that the correcting model does not verify whether the changes recommended by the evaluation system may introduce inconsistencies.

The automatic validation using LLM and KG is limited to the data from the MSD Manuals as a medical reference. Therefore, in the future, it would be beneficial to explore other reference sources, such as national medical organizations or health authorities' guidelines, in the process of evaluating discharge summaries.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT (GPT-4o) to improve language and readability and provide suggestions for better phrasing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Hyperparameters

We used the following hyperparameter values for the synthetic discharge summary generation task: temperature = 0.7, topP = 1, frequency_penalty = 0, presence_penalty = 0 for GPT-4-0125-preview and 1 for Llama 3, max_tokens = 4096. The values for top P, frequency_penalty, and presence_penalty are the default ones for the respective model. A higher temperature value allows for the generation of more diverse medical summaries. In experiments with low-temperature values (0.1–0.6), the generated summaries tended to be very similar. However, higher temperature values (0.8–1) often result in incoherent texts.

When summarizing the medical reference, we used the same values for all hyperparameters as in the generation step, except for temperature, which was set to 0.3, as Van Veen et al. showed that lower values lead to better summarization results in terms of BLEU, ROUGE and MEDCON scores [44].

For the knowledge graph generation and discharge summary correction, we used GPT-4o-2024-08-06 with the following parameters: temperature = 0.1, top_p = 1.

Appendix B. Human expert evaluation

B.1. Annotation guidelines categories

- Accuracy — does the information in the synthetic discharge summary align with established facts about the disease and the widely accepted diagnostic and treatment practices described in the MSD manual for the specific condition?
- Completeness — does the synthetic discharge summary omit crucial information to understand the patient's case and/or monitor it over time? This includes missing tests and procedures or incomplete recommendations for rehabilitation, diet, and follow-up visits.

- Redundancy — does the synthetic discharge summary contain information that is irrelevant in the context of the document or makes interpreting the discharge summary difficult?
- Style — is the information in the synthetic discharge summary presented clearly? Are the standards for medical documentation followed? Is the language appropriate?

Code and data availability

The code and dataset for this study are publicly available on the Zenodo https://zenodo.org/records/17127130.

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